Technische Universität Dresden

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Prof. Dr. Raimund Dachselt

Data Visualization

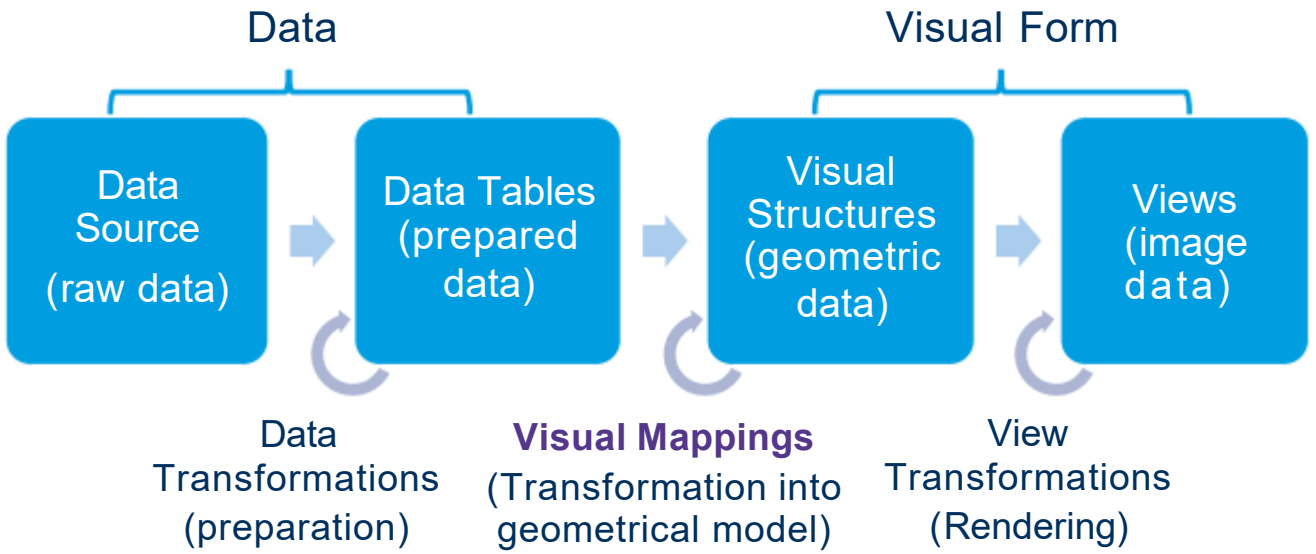
3 – Mapping to Visual Variables

Computergraphik und Visualisierung

INTERACTIVE MEDIA LAB DRESDEN

Faculty of Computer Science – Institute of Software and Multimedia Technology

Visual Variables
Goals of Visual Mapping



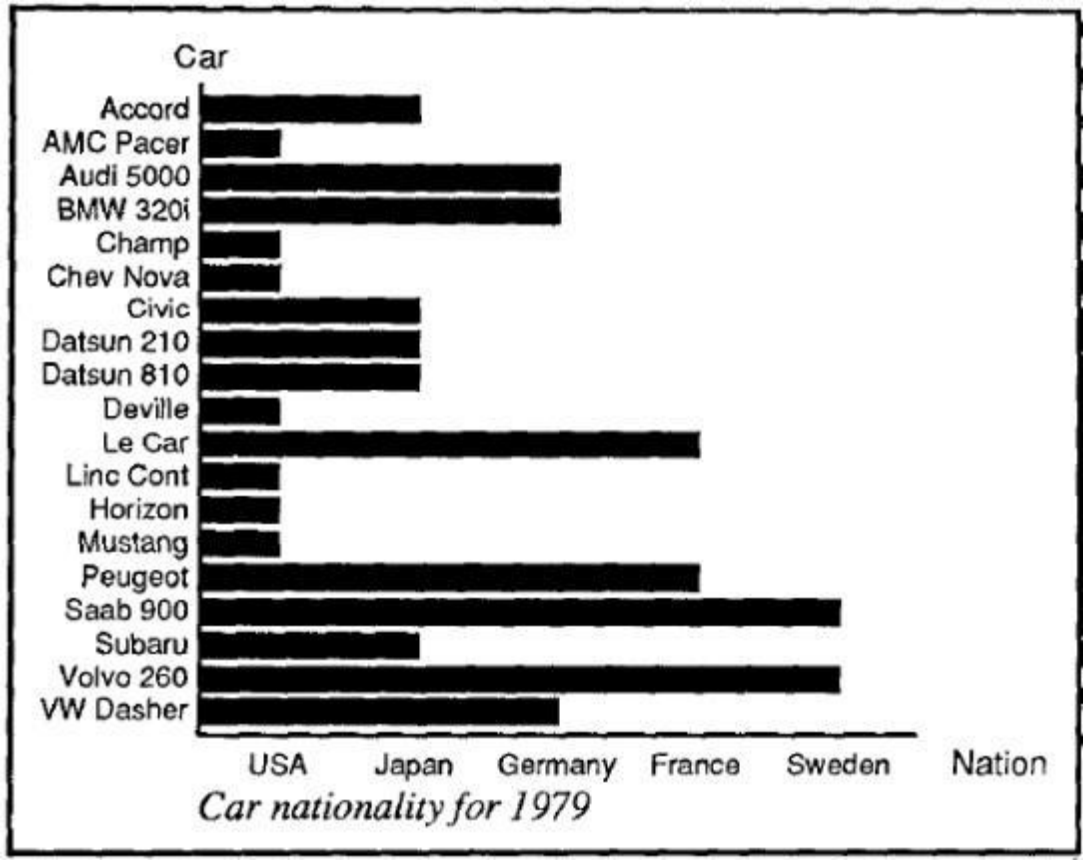
The core of Data Visualization is the Mapping to Visual Structures, what decides whether a visualization is **good or bad**.

Design Goals

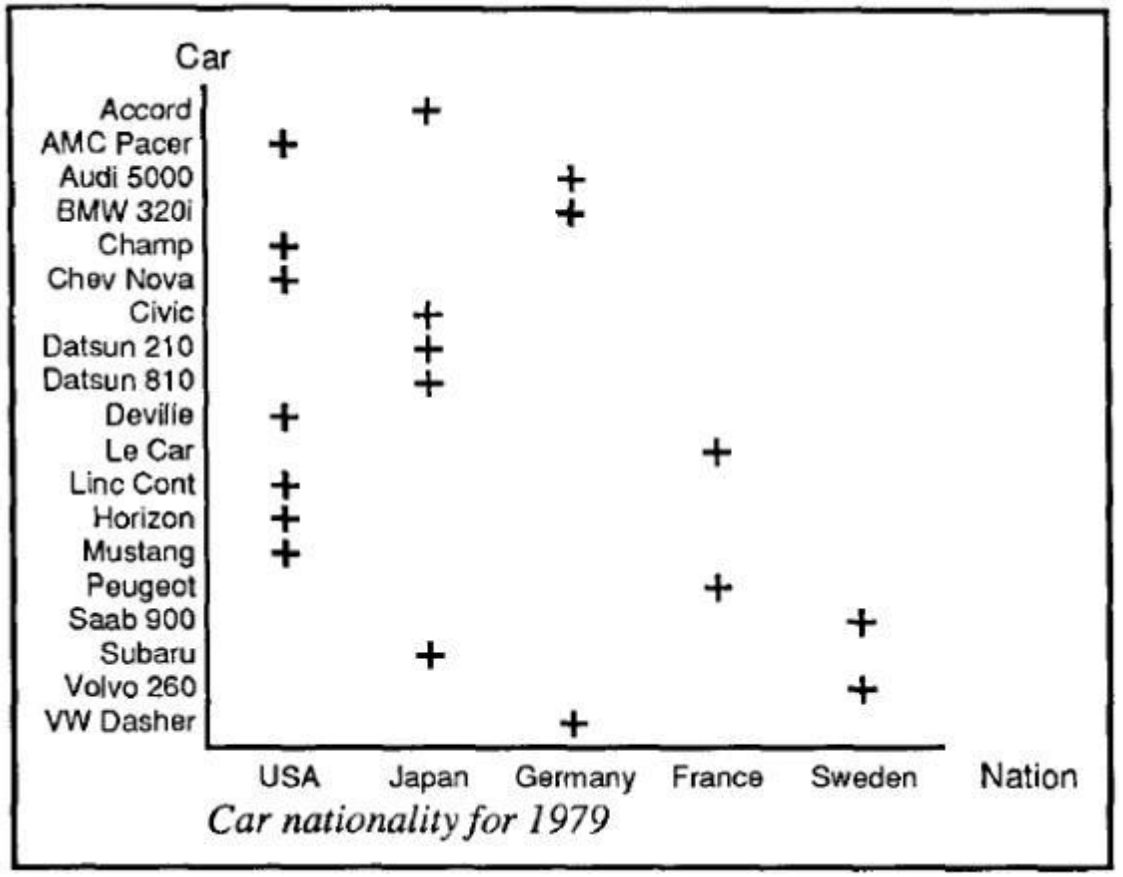
- Effectiveness: observer can perceive and comprehend information fast
- Expressiveness: visualization shows exactly the data and **nothing more**

Visual Variables

Expressiveness



not expressive – the length ofthe bars visually brings the different nations into an order, which isnot an information ofthe data [Macki`86]



expressive – noadditional information shown [Macki`86]

Visual Variables
Lecture Outline

- Semiology of Graphics
- Effectiveness Maximization
- Mapping to Color
- Mapping to Texture
- Summary

Mapping to Visual Variables

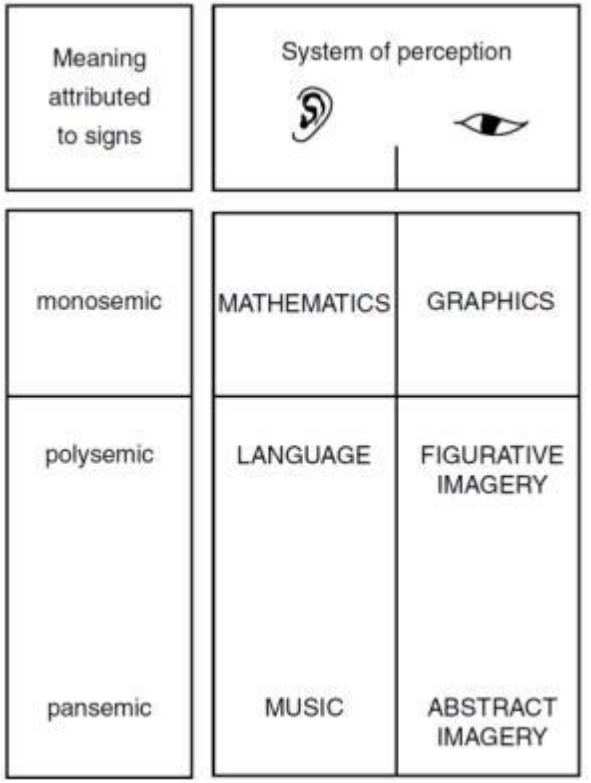
(1) Semiology of Graphics

Semiology of Graphics
Foundations

- In 1983 University of Wisconsin Press published an English translation of Jacques Bertin's seminal work **Sémiologie graphique** from **1967** with detailed analysis of 2D graphical representation of data (invariants, data dimensions (components), categories, number of elements ...)
- **Semiology** is the study of signs in order to translate information into symbols
- Example: *'On July 8, 1964, stock X on the Paris exchange is quoted at 128 francs; on July 9, it is quoted at 135 francs'*
- Information can often be captured in a table:

Stock X	
date	**quote**
7/8/1964	128
7/9/1964	135

- We aim for a monosemic encoding, where each symbol has a single meaning, as opposed to polysemic encodings where the meaning of a symbol depends on the context



Graphics in relation to other basic sign systems. [Bertin`11]

[Bertin`83] … Bertin, J.: *Semiology of graphics* (Vol. 3). Madison: University of Wisconsin press, 1983.

[Bertin`86] … Bertin, J.: *Graphische Semiologie: Diagramme, Netze, Karten,* de Gruyter, 1986 (SLUB – Drehpunkt: ZI 9710 B544S4)

[Bertin`11] … Bertin, J.: "General theory, from semiology of graphics." Cha.2 in The Map Reader, (2011): 8-16.

Semiology of Graphics
Graphical system of signs

- Bertin focuses on graphics printed on paper (no motion, no 3D)
- As graphical signs he uses **marks** drawn in plane.
- Bertin distinguishes 3 mark types:

Point (a location)

- "A point represents a location on the plane that has no theoretical length or area. This signification is independent of the size and character of the mark which renders it visible."
- marks that indicate points can vary in all visual variables

Line (a boundary, a route, a connection)

- "A line signifies a phenomenon on the plane which has measurable length but no area. This signification is independent of the width and characteristics of the mark which renders it visible."

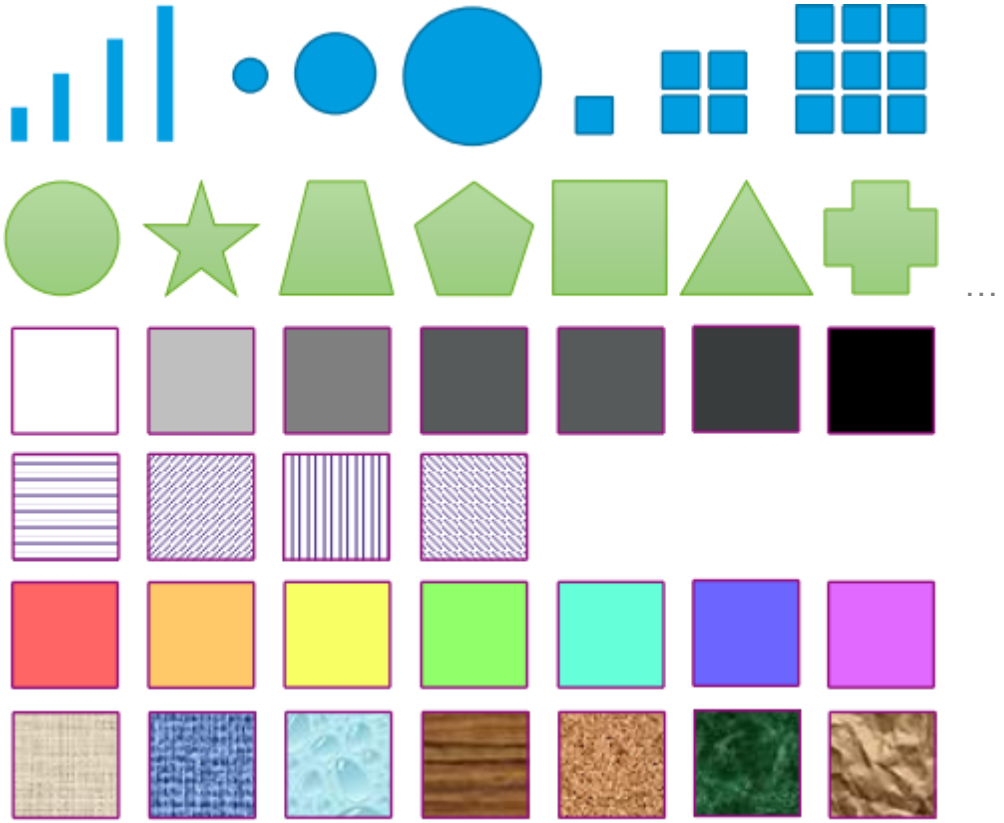
Area (a city, a country)

- "An area signifies something on the plane that has measurable size. This signification applies to the entire area covered by the visible mark."
- an area can change in position but not in size, shape or orientation without making the area itself have a different meaning

Semiology of Graphics
Visual Variables

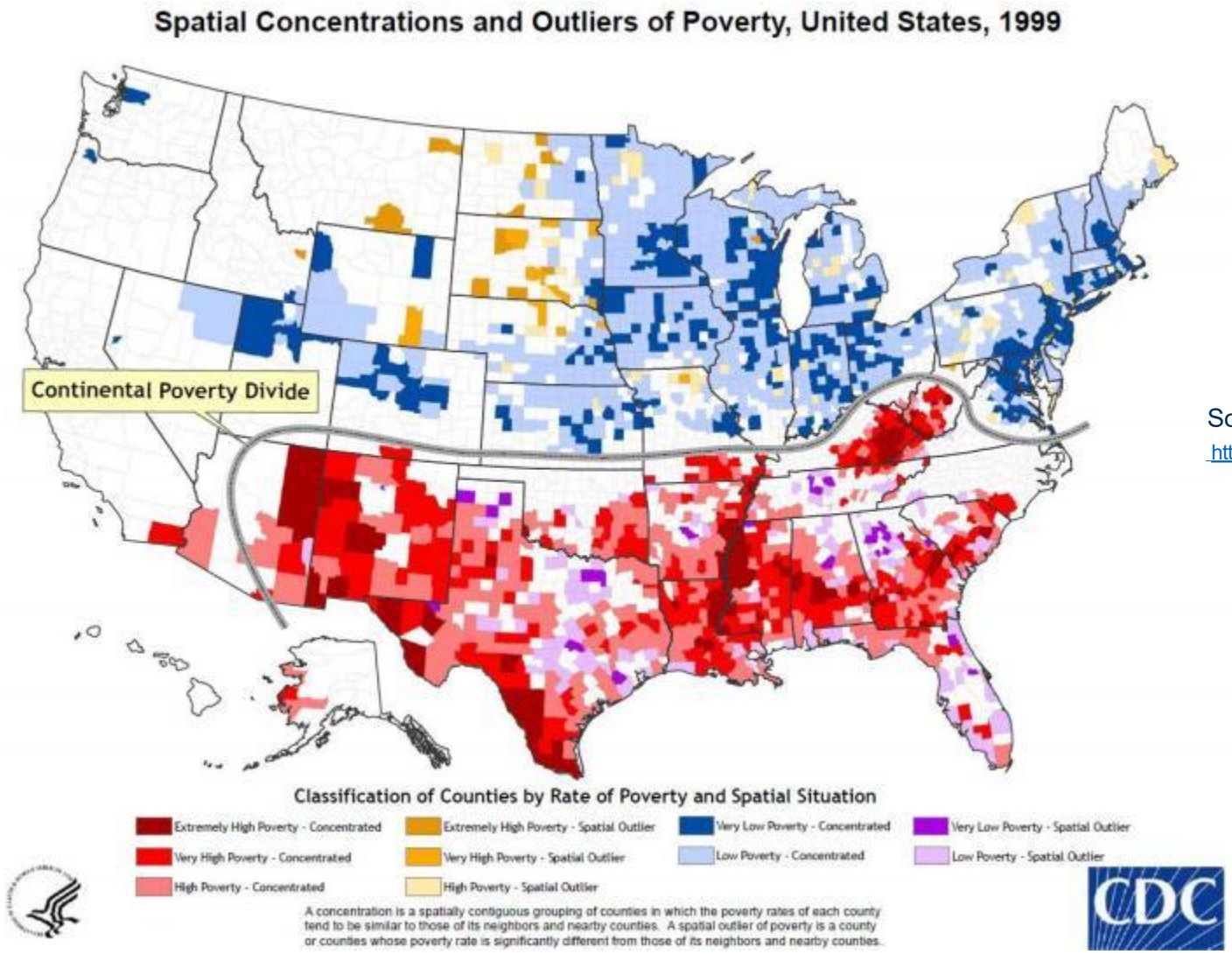
Bertin uses **8 visual variables** split into **2 dimensions** (x,y[,z] Position) and **6 retinal variables**, which are necessary as soon as we need to encode >2 features:

- size
  - change in length, area or repetition
- shape
  - infinite number of shapes
- value
  - changes from light to dark
- orientation
  - changes in alignment
- color
  - changes in hue at a given value
- texture
  - variation in pattern



Semiology of Graphics

example for selective and associative perception



Source:
https://www.cdc.gov/heart-disease-and-stroke-data/data-vis/us-concentrated-poverty.html

Semiology of Graphics
Perceptual Properties of Visual Variables

Bertin examines with user studies how changes in visual variables are perceived and distinguishes four perceptual types plus the length:

associative

- Different symbols are perceived similarly. Perception of structure in symbols not influenced by change in visual variable.

selective

- Different symbols are perceived differently. Different structures can be perceived for symbols with different values for the visual variable. (Perception can be associative **and** selective)

order

- Symbols with different values for visual variable are perceived in order.

quantitative

- Changes in visual variable can be read numerically.

length

- across how many changes in this variable are distinctions perceptible?

Semiology of Graphics
Perceptual User Studies

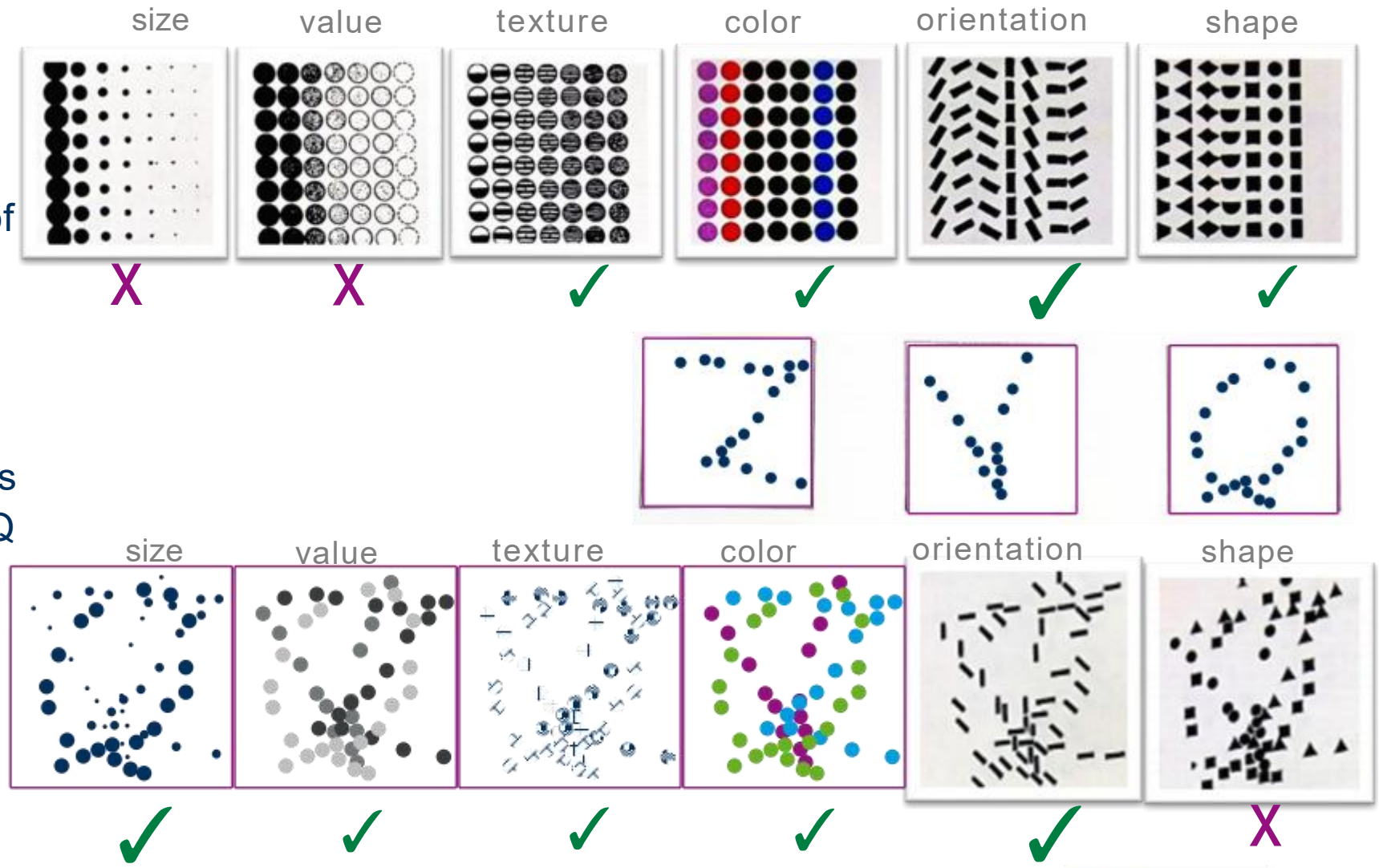
Figures taken from [Bertin`86]

Associative perception:

- show marks on grid structure
- check whether variation of visual variable does not influence perception of grid

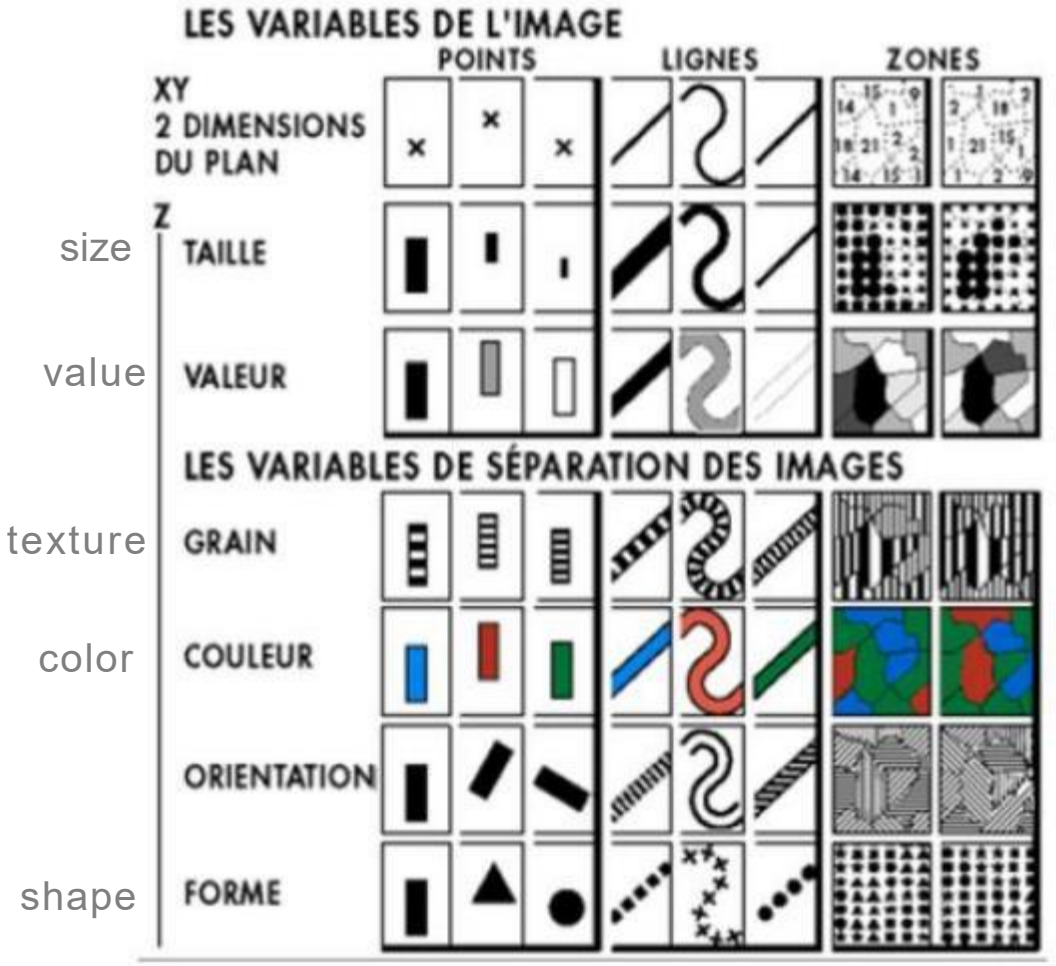
Selective Perception

- define three arrangements according to letters Z, V&Q
- check whether letters can be perceived selectively when visual attribute is varied per letter



Semiology of Graphics

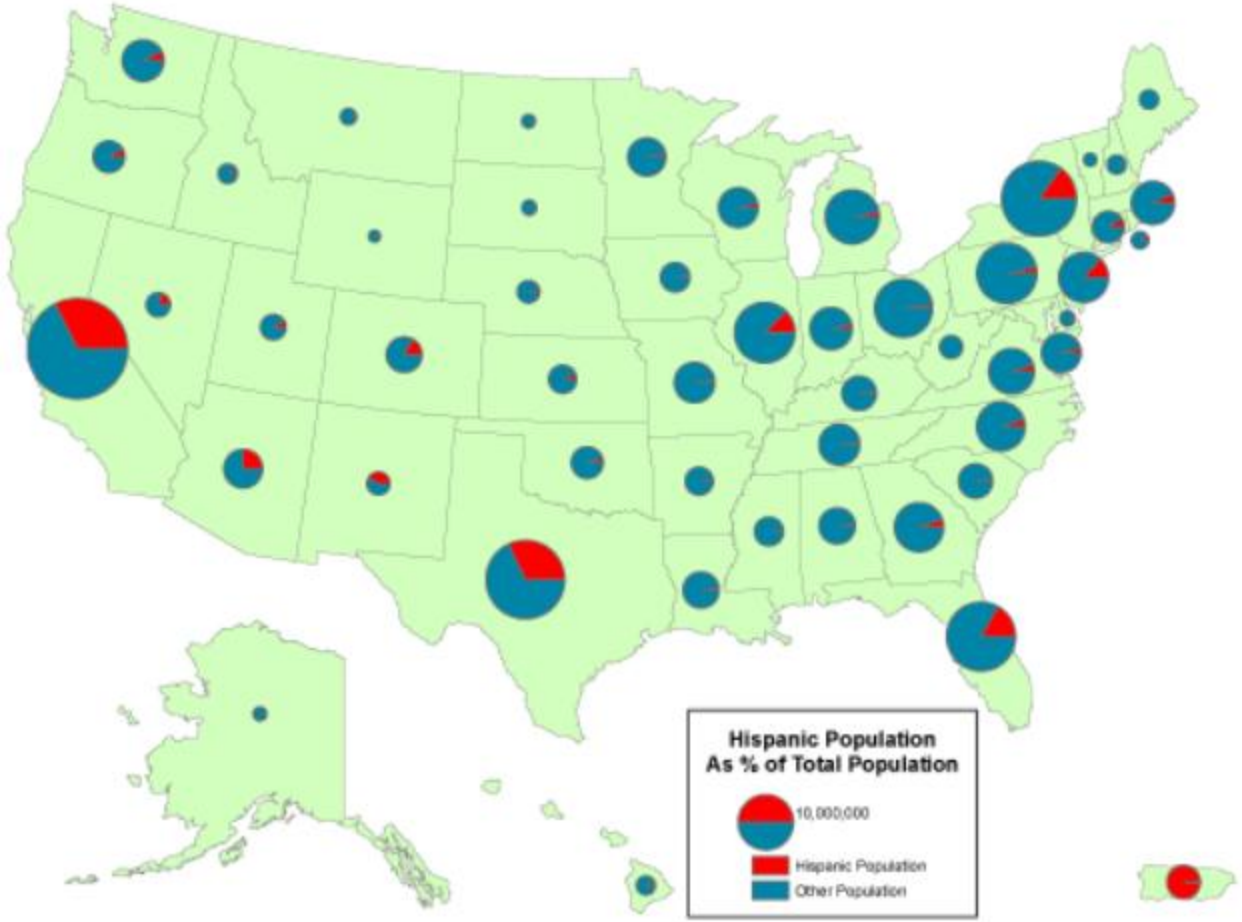
Perception of Visual Variables



[Bertin`83]

Associative	Selective	Order	Quantitative	Length
yes	yes	yes	yes	infinite
no	yes	yes	partially	select/assoc **5** distinction **20**
no	yes	yes	no	select/assoc**7** distinction **10**
yes	yes	multi	no	infinite
yes	yes	no	no	select/assoc**7** distinction **10**
yes	yes	no	no	≈5
mostly	no	no	no	infinite

Semiology of Graphics
example for quantitative perception



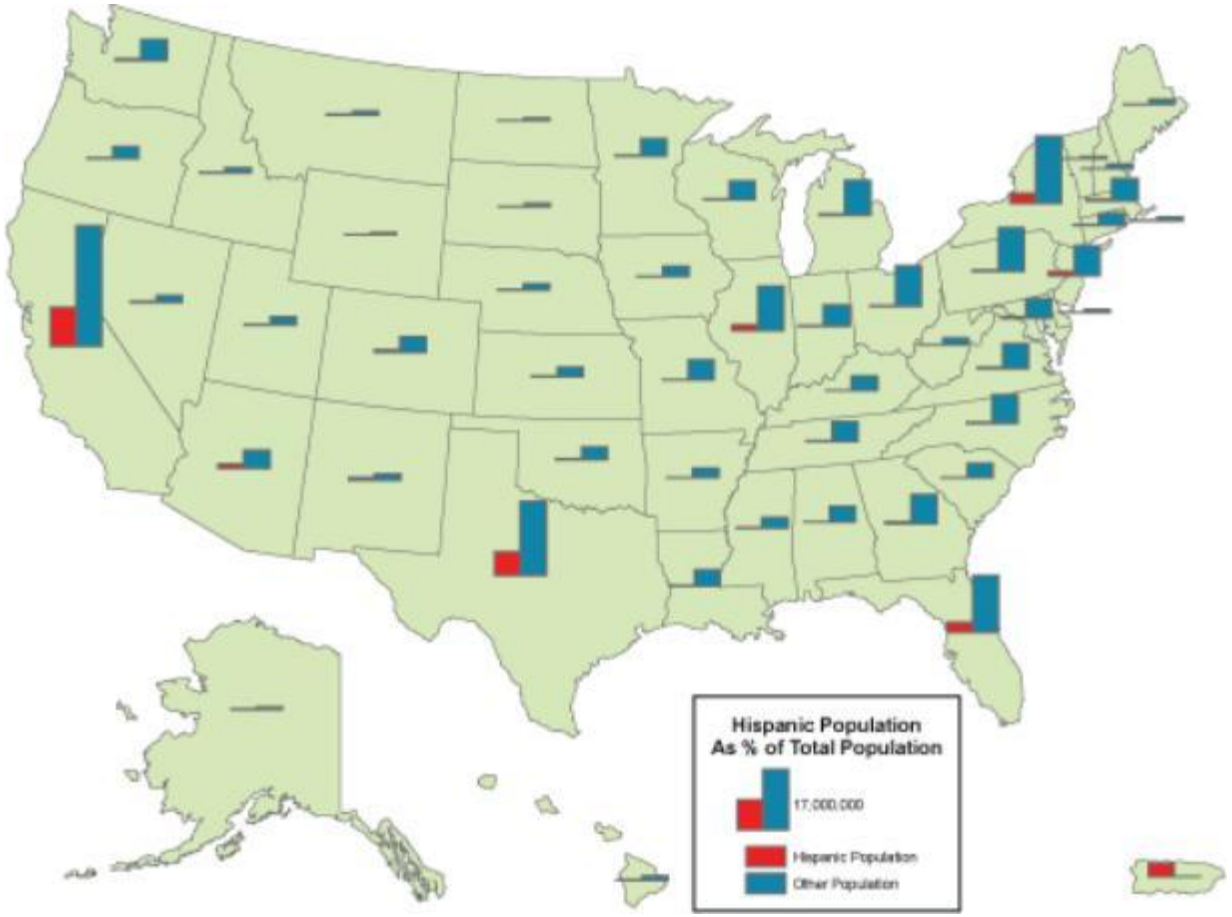


Image source: https://www.e-education.psu.edu/natureofgeoinfo/c3_p17.html

Semiology of Graphics
Mapping Goals and Guidelines

According to the perceptual type we can define mapping goals:

associative

- Look at all categories of a certain variable at the same time
- "Where is all the fish?" or "Where are all the software classes?"

selective

- Look at certain categories of a certain variable
- "Where is the salmon?" or "Where are the abstract software class?"

order

- Look at order in one dimension
- "Where does the fish taste better?" or "Which software class has more methods?"

quantitative

- Look at ratios between elements or elements of same size.
- "Where is the fish longer than 1m?" or "Is fraction of number of attributes over number of methods two or more?"

Guidelines

Position

- performs best for all goals
- use other visual attributes only if no more positional dimensions left

Color

- for associative and selective mapping

Shape

- for purely associative mapping

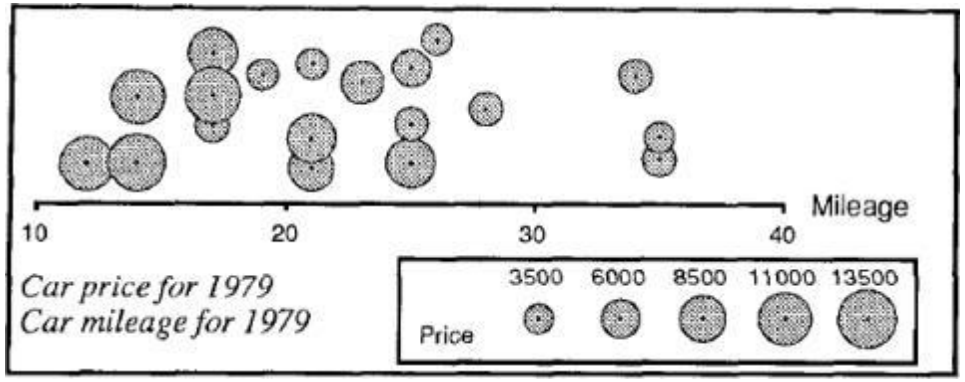
Size

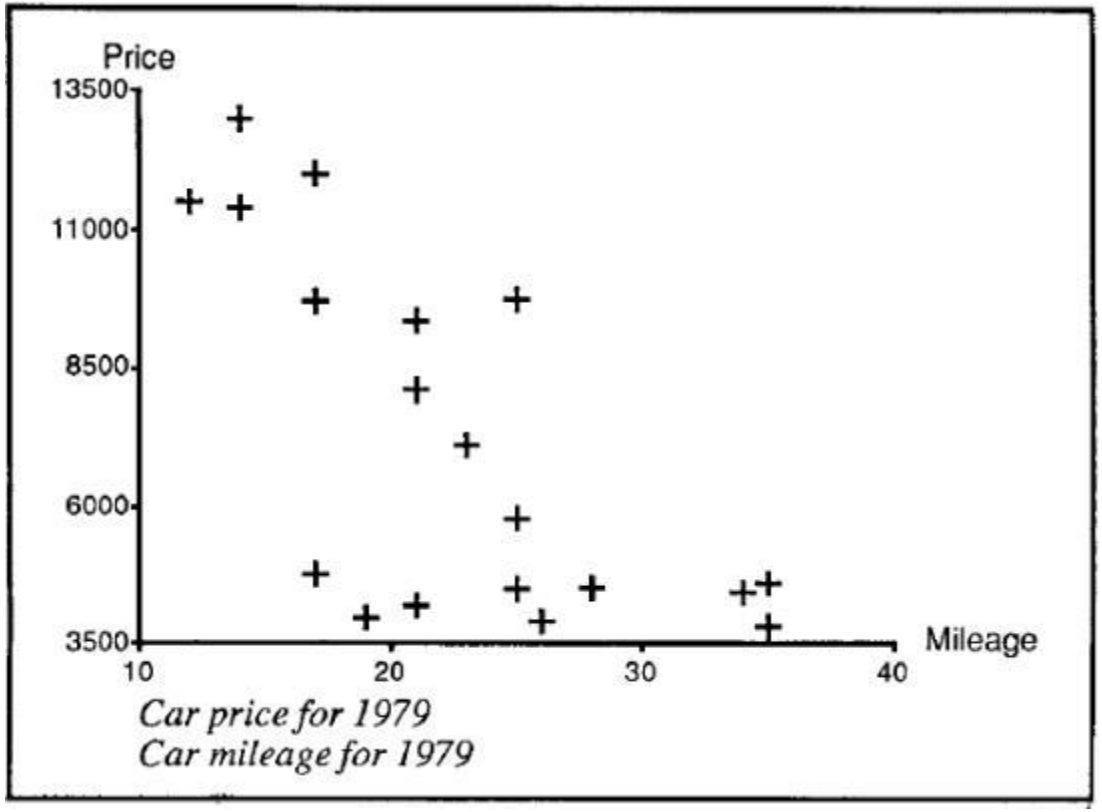
- for selective, ordered and partially also for quantitative mappings

Mapping to Visual Variables

(2) Maximizing Effectiveness

Effectiveness Maximization
Example Car Prices and Mileage

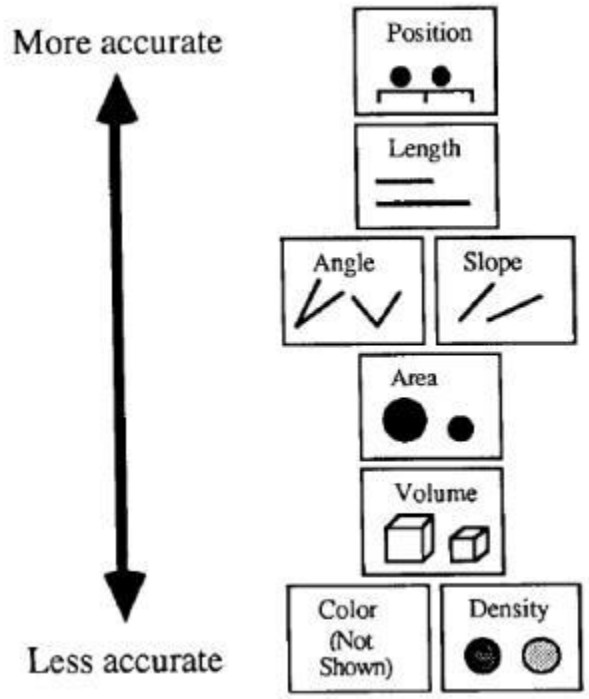




Scatterplot on the right is more effective for the perception of price [Macki`86]

Effectiveness Maximization
Mackinlay

- 1984 Cleveland & McGill performed user studies to examine effectiveness of visual variables for quantitative features.
- 1986 Mackinlay gathered more psychophysical results to rank effectiveness also for qualitative features and developed an automatic system for the generation of diagrams from a formal data description with the two selection criteria:
  - Expressivity … the diagram shows exactly the information from the data and not anymore (criterion must be met)
    (e.g. don't map nominal features to visual variables with order perception)
  - Effectiveness … speed with which the viewer can grasp the information
  - System allows additional prioritization of user specified features



Order by Effectiveness for quantitative Features [Cleve`84] illustrated in [Macki`86]

[Cleve`84] … W. S. Cleveland, R. McGill. 1984. Graphical perception: Theory, experimentation and application to the development of graphical methods. J. Am. Stat. Assoc. 79, 387 (Sept. 1984), 531-554.

[Macki`86] … Jock Mackinlay. 1986. Automating the design of graphical presentations of relational information. ACM Trans. Graph. 5, 2 (April 1986), 110-141

Effectiveness Maximization

Mackinlay's Rankings

more effective



less effective

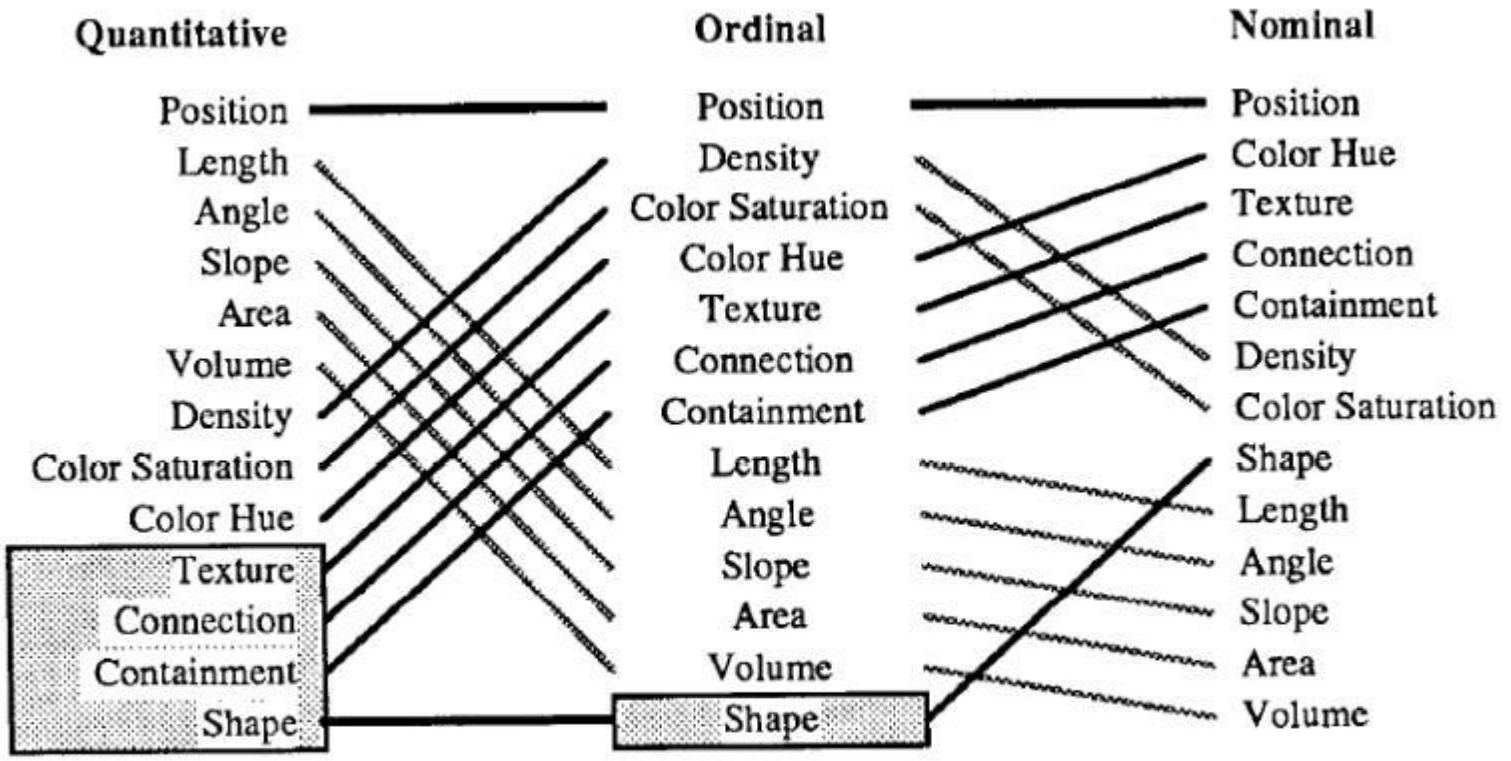


Fig. 15. Ranking of perceptual tasks. The tasks shown in the gray boxes are not relevant to these types of data.

[Macki`86]

Mapping to Visual Variables

(3) Mapping to Color

Mapping to Color
Fields of Application

Fields of Application for color:

- Distinction and grouping of elements
- Emphasis and marking of elements or areas
- Replica of natural phenomena
  - For example, Temperature (red = hot, blue = cold)
- Expression of mood

Attention!

- Interpretation of colors (e.g., mood or natural appearance) is often culturally different
- color blindness

Colour	Culture					
	Anglo-Saxon	Germanic	Latin European	Nordic	Slavic	Hispanic American
	Australia, New Zealand, UK, USA	Austria, Germany Holland, Switzerland	Belgium, France Italy, Spain	Denmark, Finland Norway, Sweden	Russia, Poland, Czech Rep., Slov., Hungary, Romania	Argentina, Chile Peru, Colombia, Mexico, Venezuela
White	Happiness, Purity, For Brides					
Blue	Dependable, High Quality, Corporate, Trustworthy and Masculine (USA)	Warmth, Feminine (Holland)		Cold, Masculine (Sweden)		
Green	Envy, Good taste, Adventure (USA); Happiness		Envy (Belgium)			
Yellow	Warmth, Happy and Pure (USA); Jealousy (UK).	Envy and Jealousy (Germany); Envy (Holland)	Infidelity (France) Envy (Italy)		Envy (Russia, Yugoslavia)	Death (Mexico)
Red	Masculinity (UK); Fear, Anger, Envy Jealousy, Love and Adventure (USA); Danger, Defiance, Hostile, Strength, Excitement, Lust	Fear, Anger, Unlucky, Jealousy (Germany)	Masculinity (France)	Positive (Denmark)	Positive (Romania); Fear, Anger, and Jealousy (Poland, Russia); Envy (Poland).	Positive (Argentina); Anger, Envy and Jealousy (Mexico)
Purple	Authority, Power, Progressive, Inexpensive (USA)				Anger, Envy and Jealousy (Poland); Envy (Russia).	Death; Anger, Envy (Mexico)
Black	Fear, Anger, Envy, Powerful, Expensive (USA); Despondency, Ceremonial	Fear, Anger and Jealousy (Germany); Despondency	Fear, Anger, Despondency		Fear, Anger, Envy (Poland, Russia); Jealousy (Russia)	Fear, Anger, Envy Jealousy (Mexico)

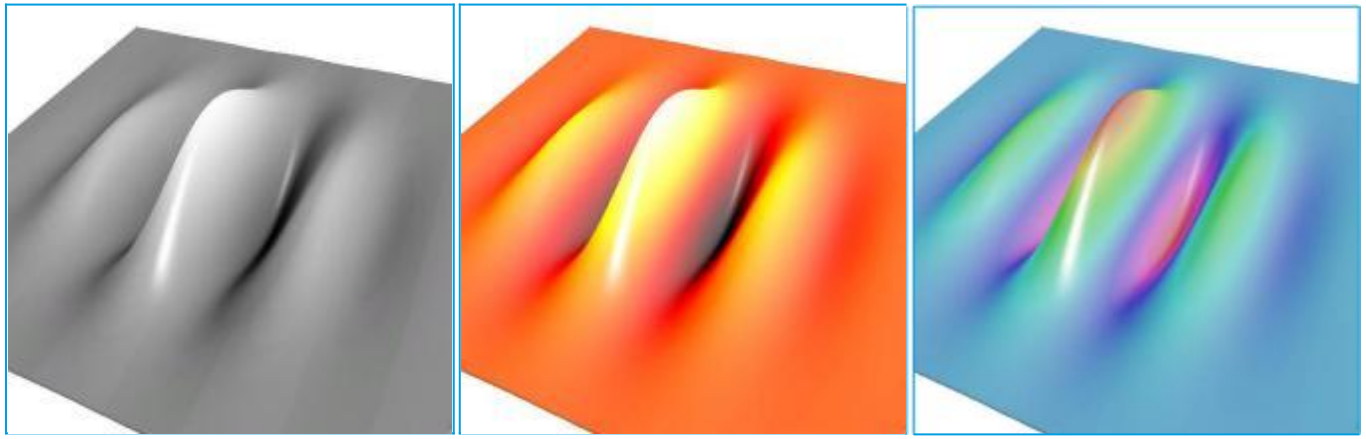
TABLE 1: THE CROSS-CULTURAL SPECTRUM OF MEANINGS AND ASSOCIATIONS OF COLOUR IN MARKETING

Aslam, M. M. (2006). Are you selling the right colour? A cross_cultural review of colour as a marketing cue. *Journal of marketing communications*, *12*(1), 15-30.

Mapping to Color
Foundations



Color used for distinction in 3D application



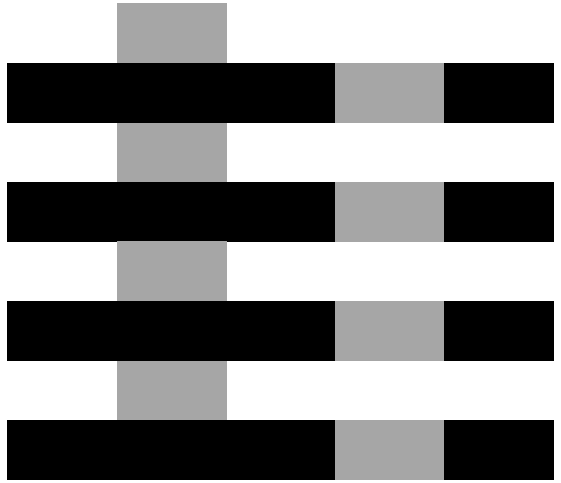
Example of Color Scales used in 3D application together with lighting

- Color differences are not perceived for tiny objects (small size color blindness)
- Minimum object size of color visualizations:
  - To discriminate colors: 16´
  - To identify colors: 20´
- measures are in arc minute which depends on viewing distance



Mapping to Color
How to describe how light something is?

- **Radiance** is light power (measured in Watt) transported along rays, i.e. power per direction (integrated to solid angle) and area orthogonal to the ray. It can be specialized per wavelength and is then called spectral radiance.
- **Luminance** incorporates the sensitivity of the human eye by integrating spectral radiance weighted with the spectral sensitivity over all wavelengths. It is measured in candela per square meter (cd/m^2)
- **Brightness** is the absolute visual perception of luminance incorporating all perceptual effects of the human visual system. Brightness increases if the overall illumination is increased
- **Lightness**, also known as **value** or **tone**, is the relative brightness in a scene and does not change with an increase of the overall illumination. It is one of the color appearance parameters of any color appearance model. See color spaces HS**V**, H**L**S, CIE **L**ab, CIE **L**uv, CIE **L**u'v'.



White's illusion: right grey blocks have same luminance but higher brightness than left grey blocks



Hermann Grid: Brightness is halucinated in line crossings although luminance is the same

Mapping to Color
Color Palette Design

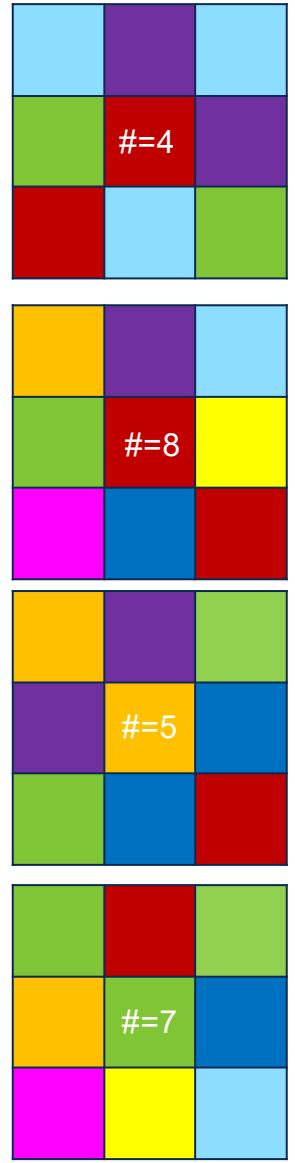
- For fast comprehension of all colors in a palette, one should restrict the total number of colors to 5-8 if possible.
- When designing a color palette the design goals are
  - maximize distinctness (perceptual distance) of colors
  - unique color names help in discussion of visualization
  - associative and selective perception
- for a **nominal feature** we additional need
  - equal lightness in order to not introduce a perceptual order

color	wavelength	color	wavelength
violett	430 nm	yellowgreen	556 nm
blue	476 nm	yellow	582 nm
greenblue	494 nm	orange	596 nm
bluegreen	504 nm	orangered	610 nm
green	515 nm	red	642 nm

10 colors that can be distinguished under most illuminations

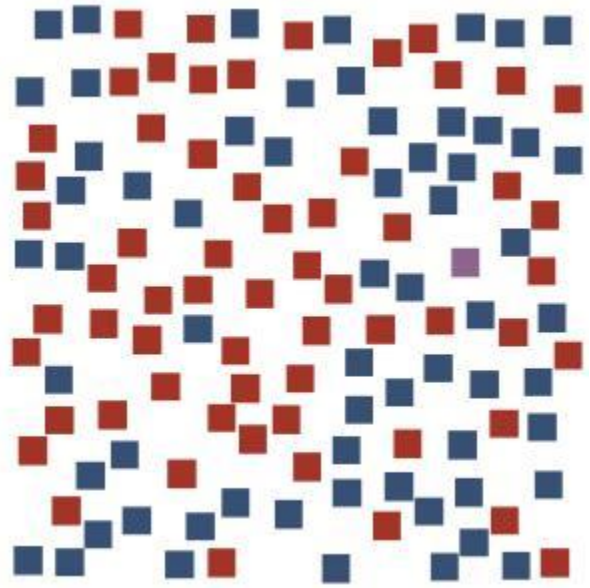
color	CIE-x-Value	CIE-y-Value
red	0,51	0,32
pink	0,35	0,30
violet	0,27	0,20
blue	0,19	0,25
green	0,31	0,52
yellow	0,46	0,48
orange	0,54	0,40
brown	0,46	0,38

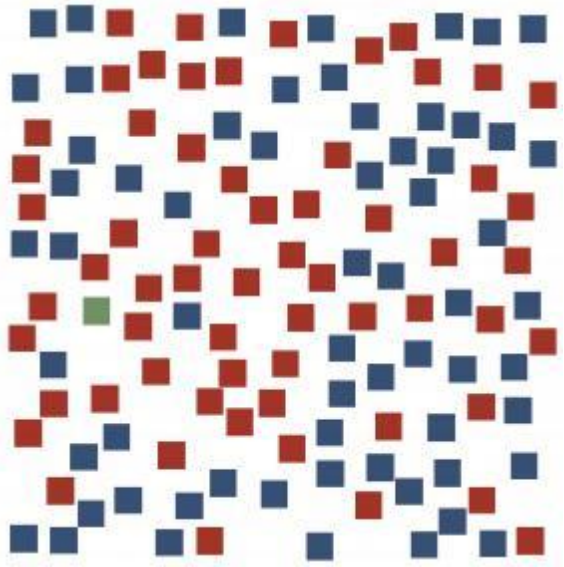
8 colors that can be named easily



Mapping to Color
Linear Separation

- Linear Separation: Preattentive selective perception for 3 colors is given only if third color is not on linear connection between two other colors in perceptual color space
- Experiment to test linear separation measures time to find target of third color:





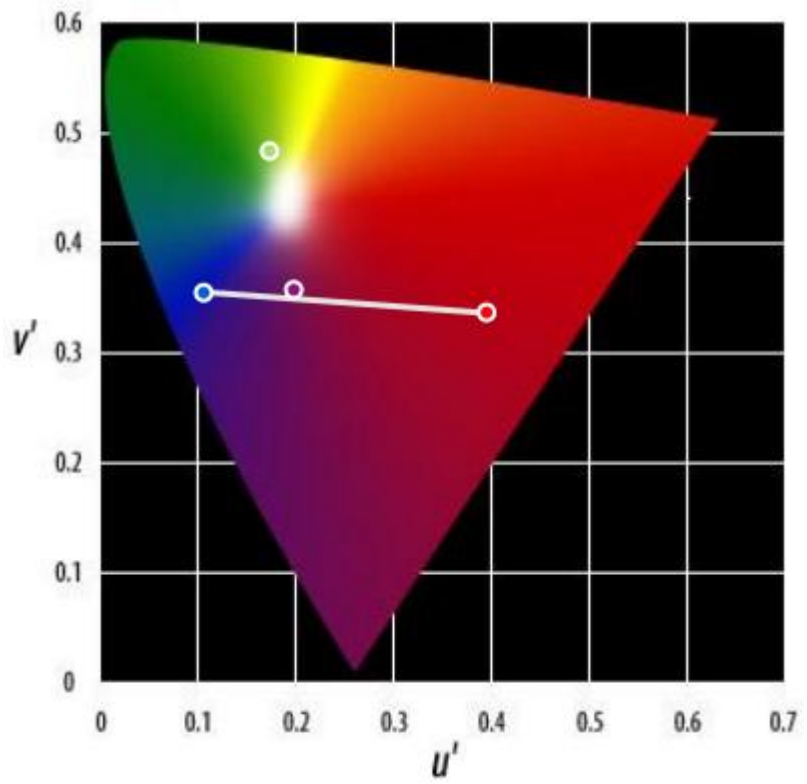
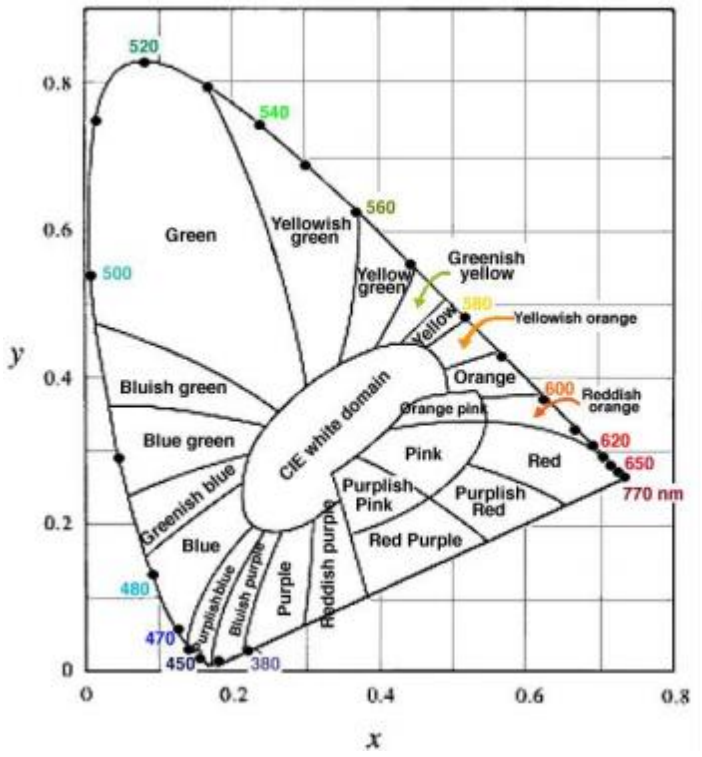


Illustration of linear separation which is not given for purple but green color with respect to blue and red

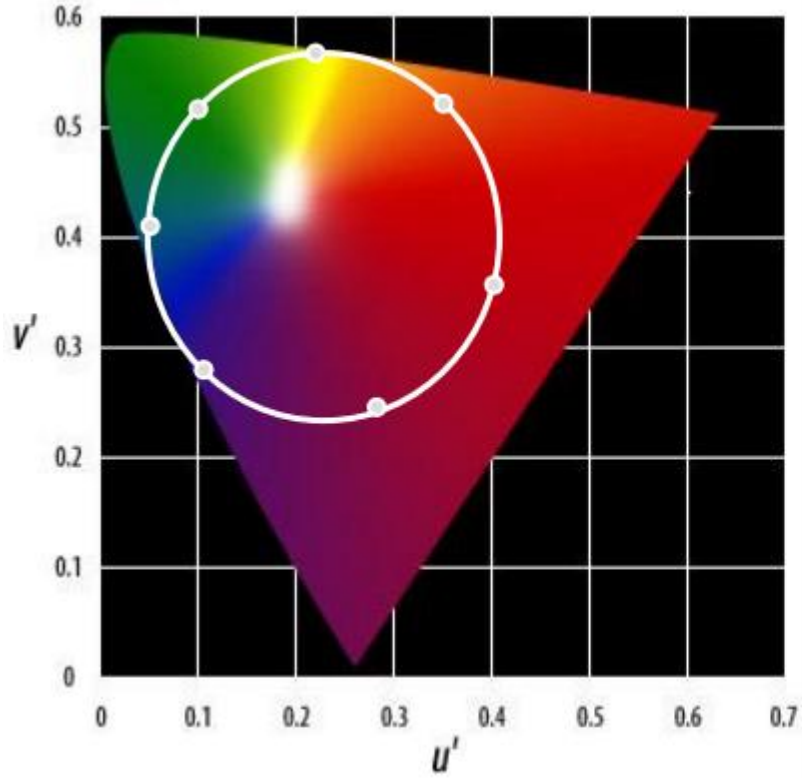
Mapping to Color
Color Palette Design for Nominal Data

Distinguishability by Maximizing Linear Separation and Choice of Different Color Categories:

- Select colors on circle in Lu‘v‘ color space so that they fall into as different categories or onto category boundaries as possible.



color regions on X+Y+Z=1 slice through XYZ color space



Equal brightness slice through Lu’v’ space

Lu’v’ is a color space where distances in Lu’v’ vectors correspond to perceived color distances

Fixing an L value yields an equal lightness color palette.

Mapping to Color
Equal Brightness Color Palette Design

- Most display systems are not calibrated for a standardized color space
- Equal brightness colors are not reproduced with equal brightness
- Kindlmann et al. proposed to exploit the face recognition for brightness adjustment:
  - generate two color image of face and show side by side with its negative
  - choose grey value with reference lightness and show it on image background
  - for each color in palette map it to foreground and adjust lightness to point where perceived face jumps from lift side to right side



face on the right side: increase lightness

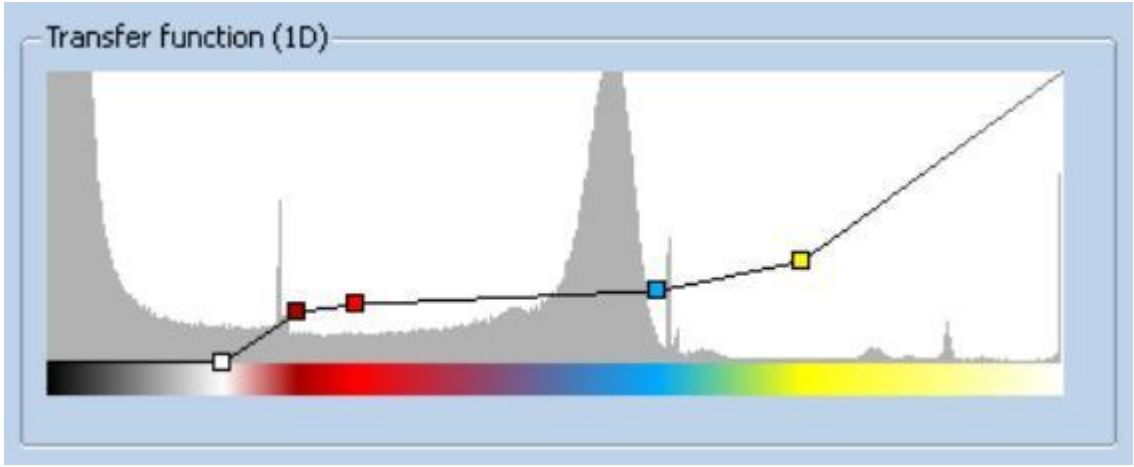


face onthe left side: decrease lightness

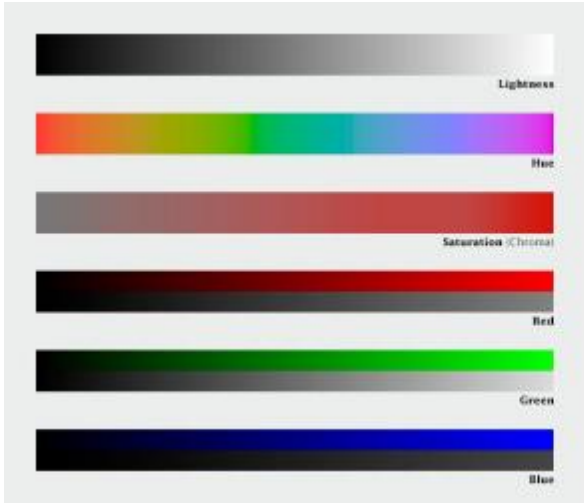
G Kindlmann, E Reinhard, and SCreem: *Face-basedLuminance MatchingforPerceptual Colormap Generation*. In IEEE Visualization, pp 299–306, 2002. (pdf)

Mapping to Color
Quantitative and Ordinal Data

- Mapping of quantitative attribute to color is called transfer function or color scale
- **Transfer functions** are typically adjustable in user interface
- **Color scales** are predefined and can be presented to the user as a choice
- One distinguishes unipolar (positive values) and bipolar (negative and positive values)
- For ordinal data color scales can be sampled equidistantly to construct a color palette



Transfer function editor for volume rendering showing attribute histogram in background



Predefined unipolar color scales



Predefined bipolar color scales

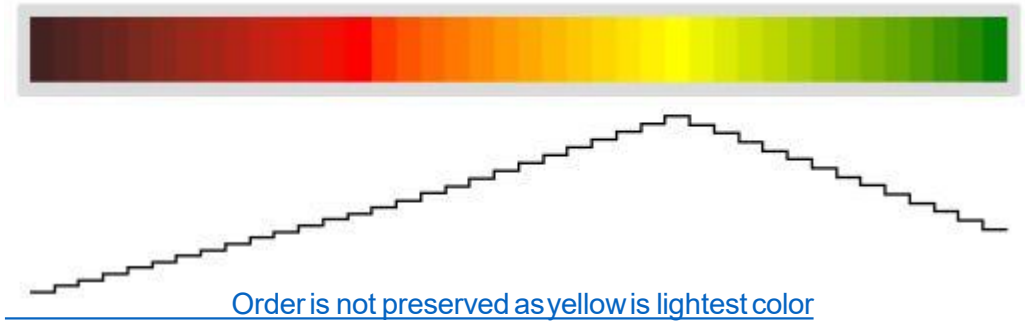
Mapping to Color
Quantitative and Ordinal Data

Design criteria

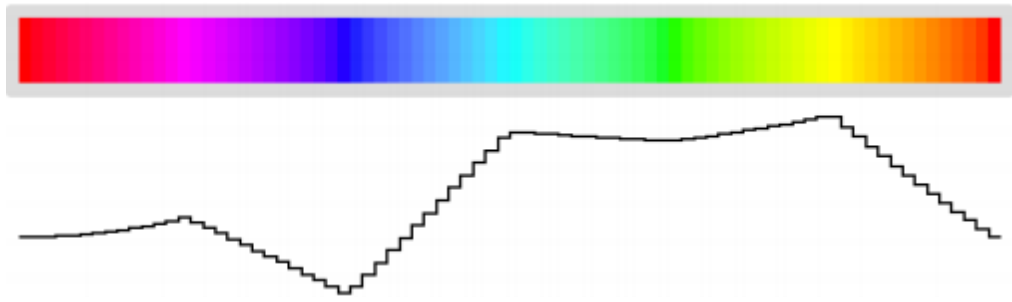
- preserve order and for quantitative attributes value differences in perceived colors (use Beziér interpolation and or lightness correction)
- Obvious start and end colors (no cycles)
- Obvious changes of the hue should be reserved for critical areas (e.g. change of the sign). These are preattentively recognized
- Colorblind save

Online tools

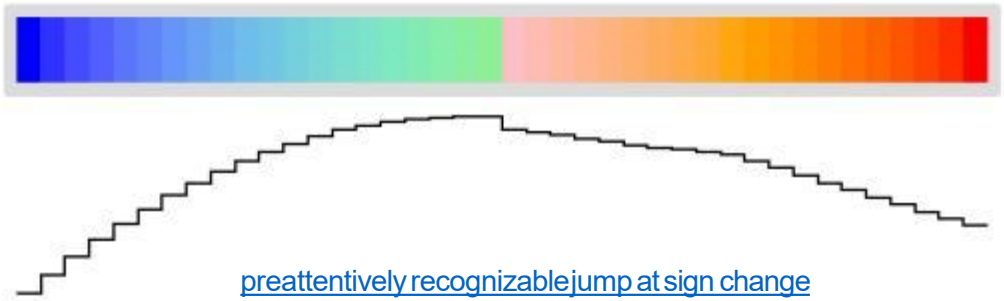
- Color Brewer
- Chroma.js



Order is not preserved as yellow is lightest color



rainbow color scale is cyclic and is not ordered



preattentively recognizable jump at sign change

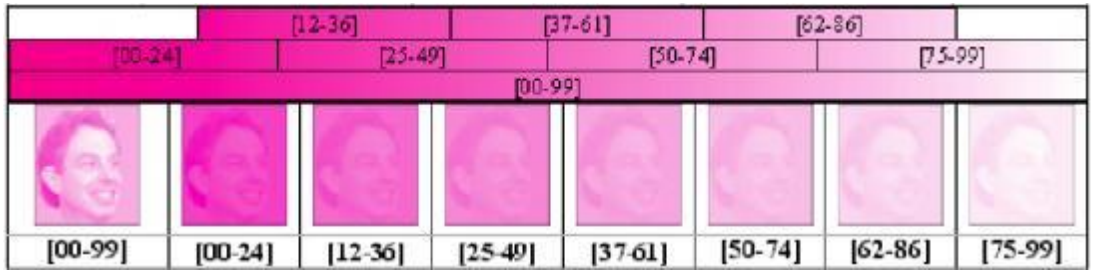
Mapping to Color
Quantitative and Ordinal Data

Evaluation of Color Scales

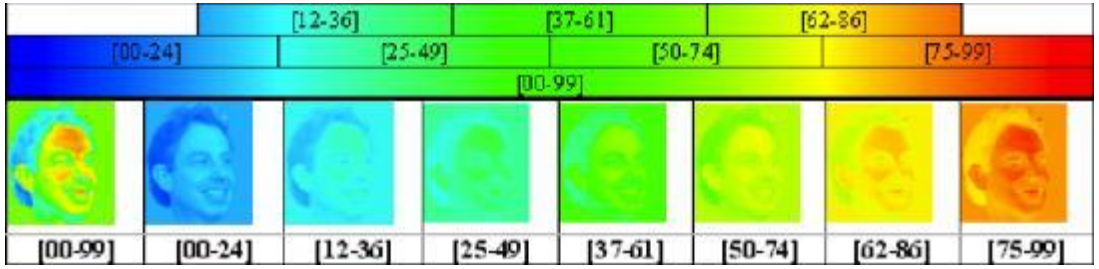
- Application to familiar, easy to interpret data
- Example: Grey values of a photography of faces.
- Motivation: People are very sensitive for lightness differences in faces.
- Specific example: “WhichBlair”-Project, Rogovitz and Kalvin, IEEE Visualization 2001, pp. 183-190
- Survey of 9 users with normal color vision with respect to the appropriateness of color scales. 7 different (overlapping) regions of the range were considered.
- Result: Strong correlation between lightness contrasts and perceived appropriateness of a color scale.



Greyscale: relatively good differentiation in all regions



Scale with **decreasing saturation**: well in upper areas



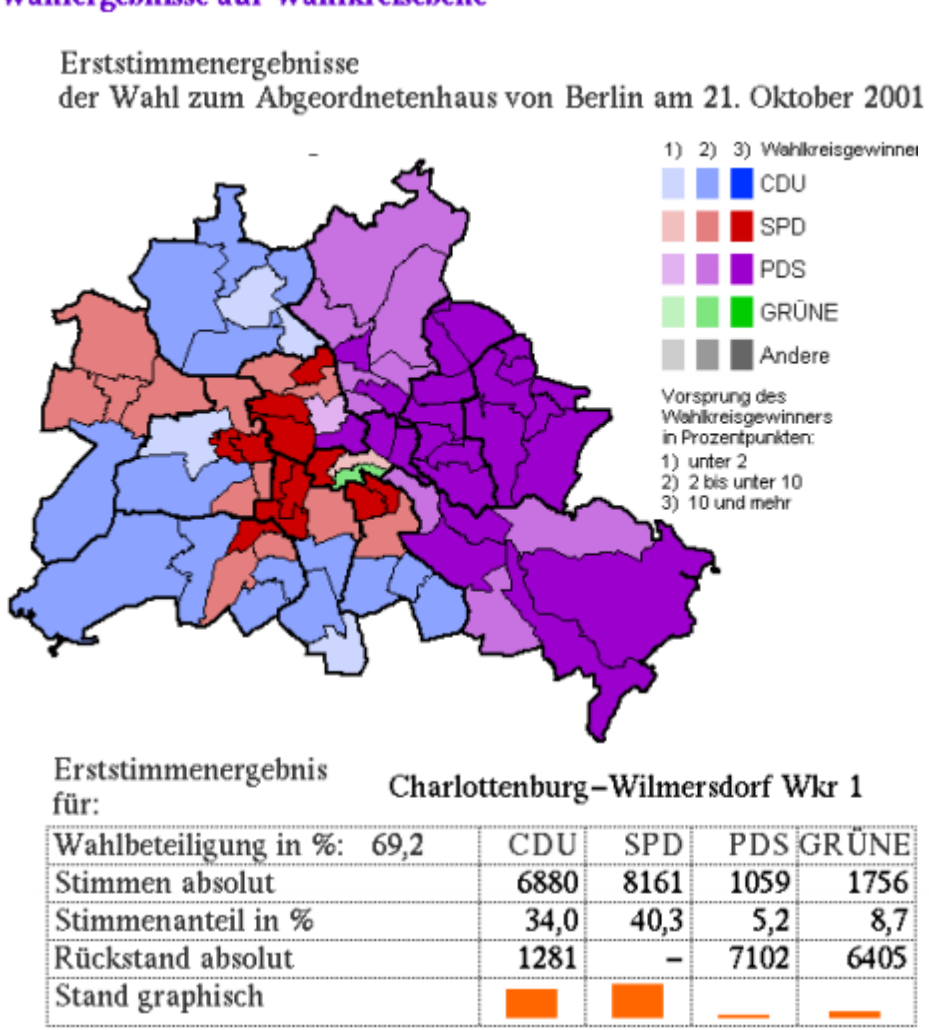
Rainbowscale: only in the lowest ¼ portion well. In several regions unnatural. Poor contrast.

Mapping to Color
Bi-variate color scales

- Mapping of 2 attributes to color
- Simple variant: Use 2 components of the RGB space ➔ not adapted to perception
- Better variants for quantitative data:
  - Use 2 components of the HSV space (e.g. Hue and Saturation)
  - Employ a scale from yellow to blue and a scale from yellow to red
- Nominal data:
  - Use strongly different colors, Maximum: 4x4- or 3x5 colors to support preattentive perception



Bad example of mapping2 attributestored and blue



Hue: winning party, **Saturation**: winningmargin

Mapping to Color
Bi-variate color scales – Design Tools and Evaluation

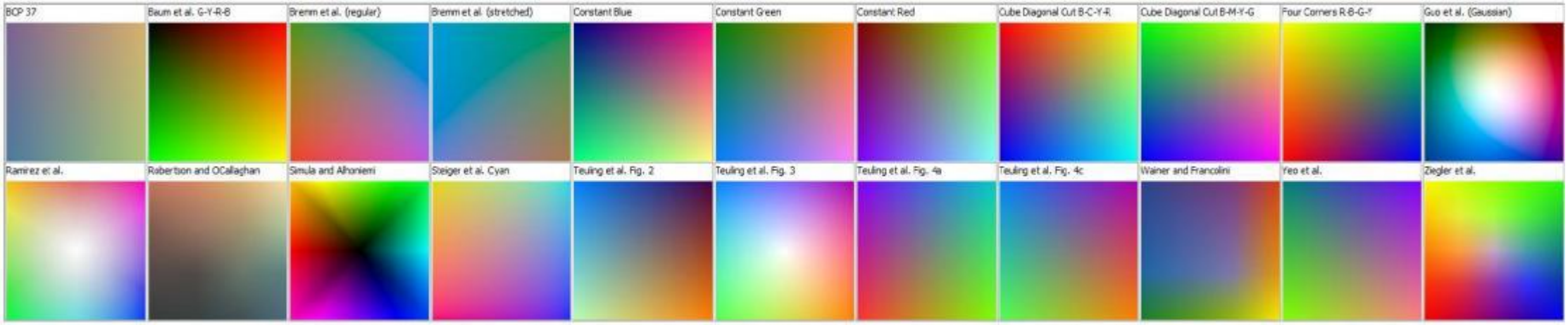


Figure 2: The initial overview lists all available color map implementations.

Image from: Steiger, M., Bernard, J., Thum, S., Mittelstädt, S., Hutter, M., Keim, D., &Kohlhammer, J. (2015). Explorative analysis of2d color maps. Václav Skala-UNION Agency.

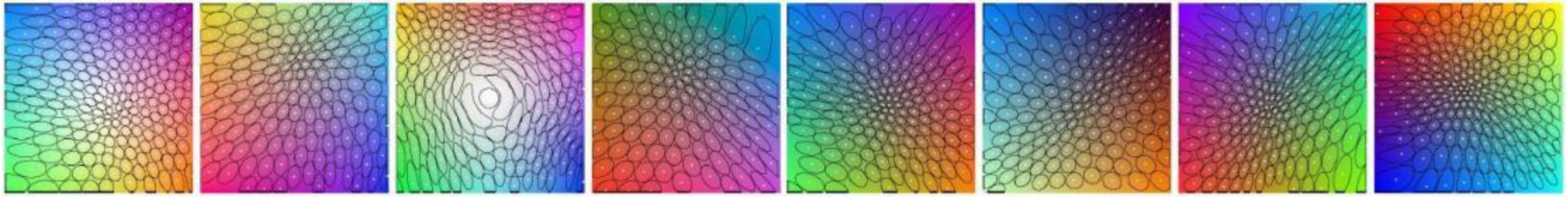


Figure 1. Static 2D colormaps with the highest overall rating for black or white backgrounds. From left to right: TeulingFig3,[1] Mittelstädt et al.,[2] Ramirez et al..[3] Bremm et al. (regular),[4] TeulingFig3NoWhitening,[1] TeulingFig2,[1] TeulingFig4a,[1] Cube Diagonal Cut B-C-Y-R,[4–8]

Image from: Bernard, J., Steiger, M., Mittelstädt, S., Thum, S., Keim, D., &Kohlhammer, J. (2015, February). Asurvey and task-based quality assessment of static 2D colormaps. In *VisualizationandDataAnalysis2015*(Vol. 9397, p. 93970M). International Societyfor Optics and Photonics.

Mapping to Color

Bi-variate color scales – Bad and Good Examples for MotionExplorer

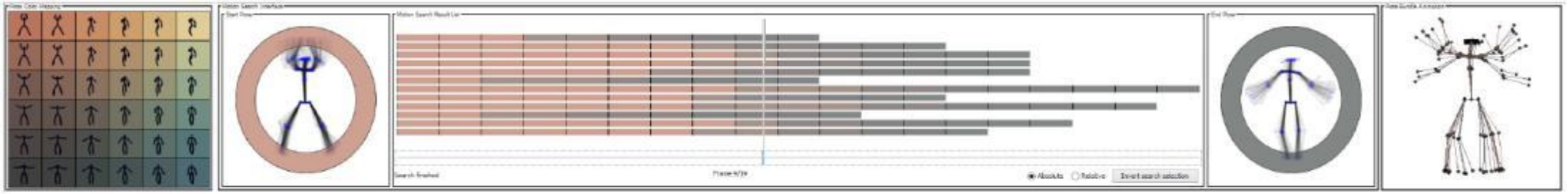


Figure 3. An inappropriate colormap for the tasks performed in MotionExplorer: Robertson and OCallaghan.

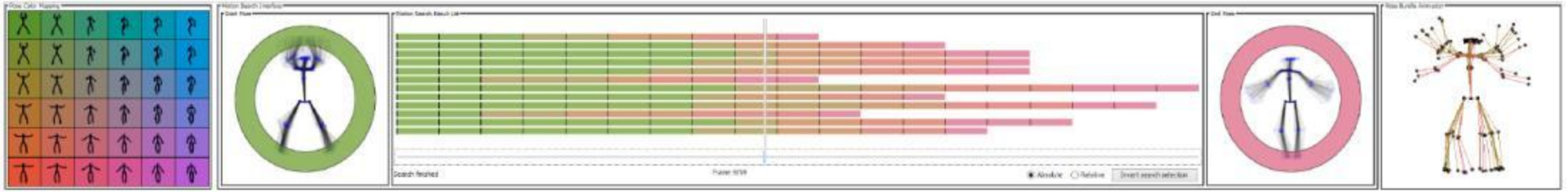


Figure 4. An appropriate colormap for the use case: Bremm et al. (regular). A sufficient number of JND Colors, distances to white and black, as well as perceptual linearity scores above average, and a good Attention Steering score.

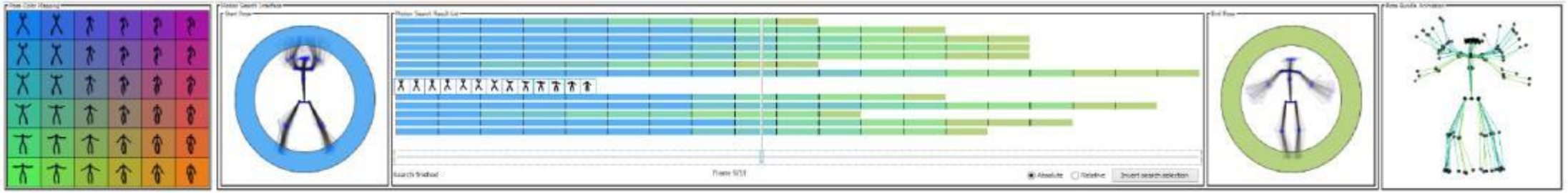


Figure 5. An appropriate colormap for the use case: Teuling3NoWhitening. More JND Colors than Bremm et al. (regular), distances to white and black, as well as perceptual linearity scores above average.

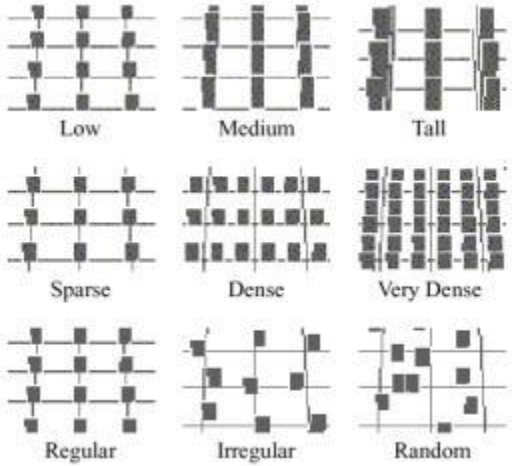
Mapping to Visual Variables

(4) Mapping to Texture

Mapping to Texture
Overview



material data base



Healey'98



Ware'95

Nominal Data

- one can use texture (2D) and material (3D) collections
- 2 nested variables can be mapped, e.g. onto category (leather, fabric, stone) and value (specific leather types)

Ordinal Data

- several texture attributes are perceived in order:
  - grain size
  - contrast
  - density
  - regularity
  - orientation

Quantitative Data

- Even though not most effective, some continuous texture attributes available:
  - grain size
  - contrast
  - orientation (cyclic value)

Mapping to Texture
Perceptual Considerations

- When textures are synthesized we need to consider perceptual limitations
- From user studies on human perception we can derive guidelines on the use of the different texture attributes :
  - maximum contrast sensitivity: between 1 and 10 cycles per degree ≈ **4-40 pixels per cycle**
  - maximum orientation resolution: approx. **5°** with higher resolution around 0 ° and 90 °
  - maximum number of distinguishable frequencies within 1 and 10 cycles per degree: **24**







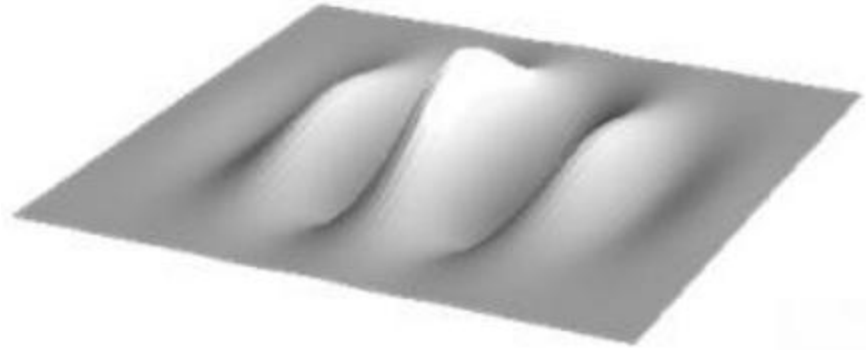
Mapping to Texture
Texture Synthesis – Gabor Functions

Ware and Knight use Gabor functions to synthesize textures with varying contrast frequency and orientation:

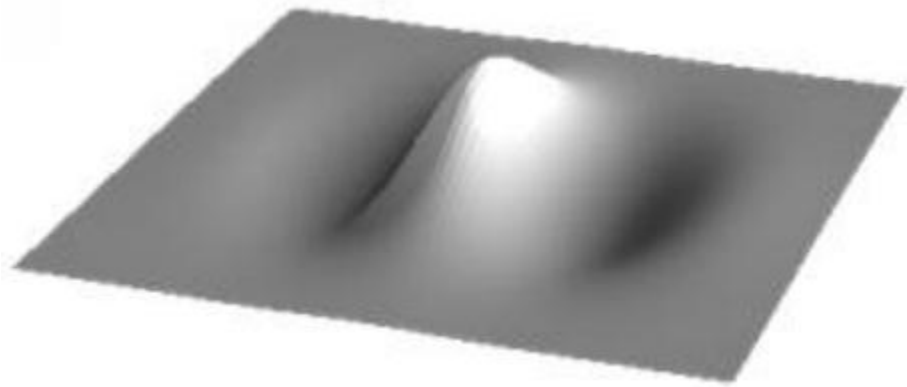
$$G_{\underline{x}_0,A,f,\hat{\underline{v}},\sigma}(\underline{x}) = A \cdot \cos\left(2\pi f \langle \hat{\underline{v}}, \underline{x} - \underline{x}_0 \rangle\right) \cdot \exp\left(-\frac{\|\underline{x} - \underline{x}_0\|^2}{2\sigma^2}\right)$$

- $\underline{x}_0$ position of Gabor function
- σ size of Gaussian window function
- A amplitude
- f frequency of vibration
- ν direction of vibration

Garbor functions are a frequently used model for the low level detectors of human vision



Examples of Gabor functions with different frequency to size ratios



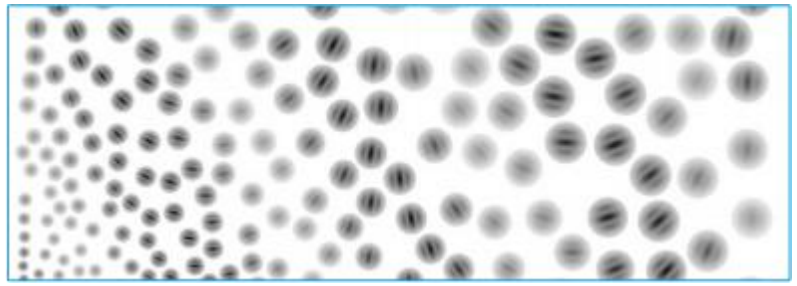
Colin Ware, William Knight: Using visual texture for information display, ACM Transactions on Graphics (TOG), 1995 (pdf)

Mapping to Texture
Texture Synthesis – Texton Mapping

Ware and Knight define a so called **Texton** t with a 2D position x_0 and 3 retinal attributes:

- contrast C … corresponds to amplitude
- size S … defines wavelength and window size
- orientation φ … corresponds to direction othogonal to vibration direction

$$t[x_0, C, S, \varphi](x) = C \cdot \cos\left(\frac{2\pi}{S}\left\langle\begin{pmatrix}\cos\varphi\\ \sin\varphi\end{pmatrix}, x - x_0\right\rangle\right) \cdot e^{\frac{1}{2}\left(\frac{\|x - x_0\|}{1.5s}\right)^2}$$



Examples of textons with varying parameters

Mapping of attributes:

Contrast: according to *Stevens' Power Law*

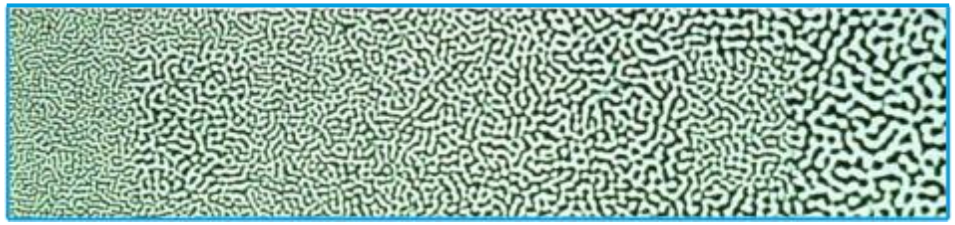
- intensity range: $0.1, 1.0$

Size: according to *Stevens' Power Law* for area

- approx. $2 __ 16$ cycles per degree
- Careful with the influence of context

Orientation: linearly in range $I_\varphi = \varphi_{min}, \varphi_{max}$

- for cyclic attributes choose: $I_\varphi = [0, 2\pi]$,
- for positive attributes choose: $I_\varphi = [0, \frac{\pi}{2}]$,
- for signed attributes choose: $I_\varphi = [-\frac{\pi}{4}, \frac{\pi}{4}]$



Examples of influence of context on size perception

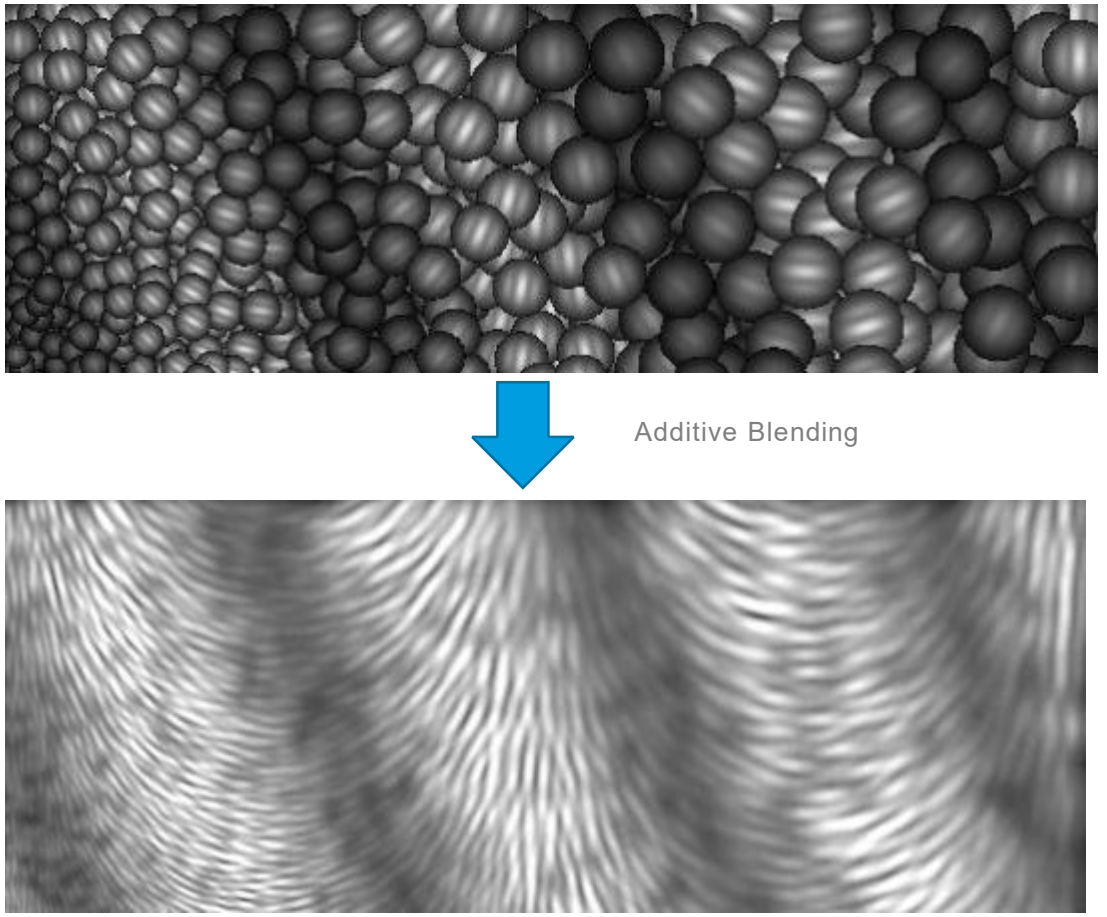
Mapping to Texture

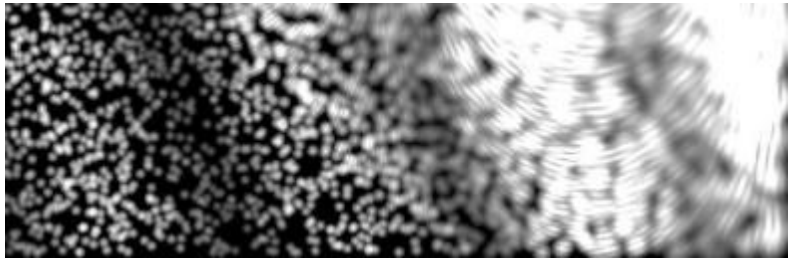
Texture Synthesis – Sampling

- To synthesize a complete texture texton locations are sampled and the resulting textons are blended with additive blending
- **Challenge:** how to sample texton locations such that we have a uniform distribution with respect to texton size?

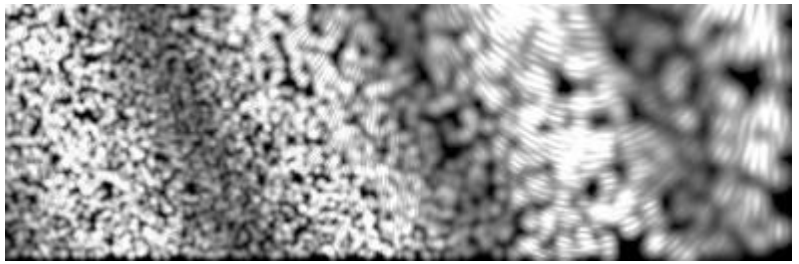
Random Sampling Approaches:

- Uniform Sampling (simple but not contrast preserving)
- Inverse Area Sampling (simple but irregular)
- Poisson Disk Sampling (awesome but more complicated to implement)

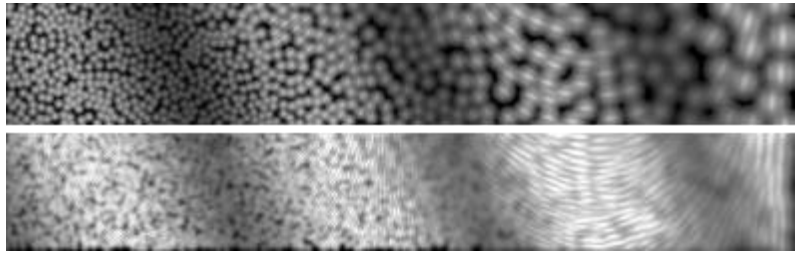




uniform sampling



inverse area sampling



Poisson disk sampling with different disk sizes

Mapping to Texture
Texture Synthesis – Uniform Sampling in C++

```
#include <random>
#include <algorithm>

void random_sample(std::vector<P2D>& P, int n) {
    std::mt19937 rand_gen;
    std::uniform_real_distribution<double> uni_dist(0.0, 1.0);
    P2D p;
    for (unsigned i=0; i<n; ++i) {
        p = P2D(uni_dist(rand_gen), uni_dist(rand_gen));
        P.push_back(p);
    }
}
```

Mapping to Texture
Texture Synthesis – Inverse Area Samping in C++

```
#include <random>
#include <algorithm>

void inv_area_sampling(std::vector<P2D>& P, int n) {
    std::mt19937 rand_gen;
    std::uniform_real_distribution<double> uni_dist(0.0, 1.0);
    P2D p;
    for (unsigned i=0; i<n; ++i) {
        do {
            p = P2D(uni_dist(rand_gen), uni_dist(rand_gen));
        } while (uni_dist(rand_gen) >
                 min_texton_area/texton_area_at(p));
        P.push_back(p);
    } }
```

Idea: accept samples with probability

$$p(x) \alpha \frac{1}{\text{area}(x)}$$

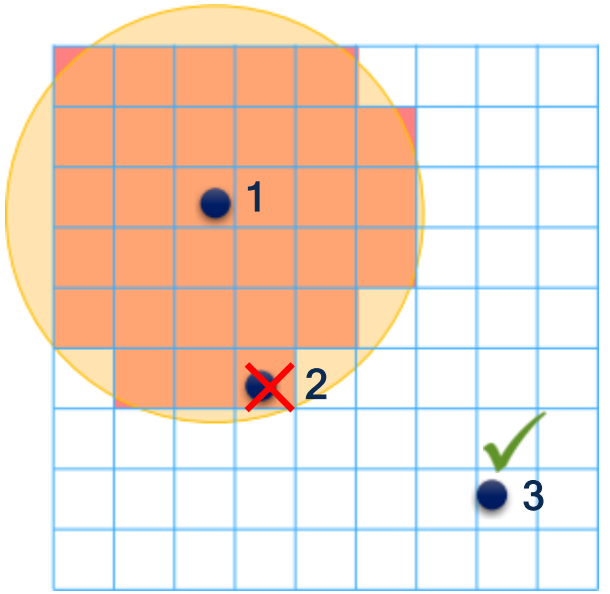
Mapping to Texture

Texture Synthesis – Poisson Disk Sampling in C++

```
#include <random>
#include <algorithm>

void poisson_sampling(std::vector<P2D>& P, int n) {
    std::mt19937 rand_gen;
    std::uniform_real_distribution<double> uni_dist(0.0, 1.0);
    P2D p;
    quadtree Q;
    for (unsigned i=0; i<n; ++i) {
        do {
            p = P2D(uni_dist(rand_gen), uni_dist(rand_gen));
        } while (Q.collides(p,poisson_scale*size_at(p)));
        P.push_back(p);
        Q.insert(p);   } }
```

Idea: reject samples too close to previous samples



For efficient implementationwe need book keepingoffreeareas

Mapping to Visual Variables

(5) Summary

Mapping to Visual Variables
Summary

You have learned …

- the most important visual variables (positional+retinal) used to encode data attributes
- The perceptual types and how to exploit them to achieve different mapping goals
- how to choose visual variables for an effective and expressive visualization based on value type and range
- the capabilities of color in visualization
- how to design a color palette or a color scale and the underlying design goals
- the capabilities of texture in visualization
- how to synthesize a texture based on textons